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Question 1 [C01] [10 Points]

In the digital kingdom of **Digitaria**, every number holds a secret code. You, the **Royal Codekeeper**, have been given a sacred task by the king to calculate the **"Vault Value"** of any given number using a mysterious rule. The rule says: For every **even digit**, add the **square** of that digit to the Vault Value. For every **odd digit less than 5**, subtract the **cube** of that digit from the Vault Value. For every digit that is **divisible by 3**, add the **digit itself** to the Vault Value. Otherwise subtract the digit to the vault Value.

The process must be done using **only one loop**. When the number has been fully processed, the **final total** you get is the **Vault Value**, the key to open the secret chamber of numbers.

Write a program that takes a number from the user and calculates the **Vault Value** according to the above magical rules. [You cannot use string, array here]

Input	Output
code_number = 46385	vault_value= 84
<u>Explanation:</u> Last digit = 5, Vault Value change: $0 - 5 = -5$ Next digit = 8, Vault Value change: $-5 + (8^2) = -5 + 64 = 59$ Next digit = 3, Vault Value change: $59 - (3*3*3) = 59 - 27 = 32$ Next digit = 6, Vault Value change: $32 + (6^2) = 32 + 36 = 68$ Next digit = 4, Vault Value change: $68 + (4^2) = 68 + 16 = 84$	